

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12402533
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Murata; Tomonori et al.

Residual charge removal device

Abstract

A residual charge removal device includes: a power storage unit; a first switching unit that is connected to a target to which electric power is supplied from a power supply device, and operates to remove a residual charge of the target by electric power supplied from the power storage unit; and a second switching unit that operates to supply electric power from the power storage unit to the first switching unit when electric power supply to the target is stopped.

Inventors:	Murata; Tomonori (Kanagawa, JP), Shibata; Isao (Kanagawa, JP)
Applicant:	KELK Ltd. (Kanagawa, JP)
Family ID:	1000008779894
Assignee:	KELK Ltd. (Kanagawa, JP)
Appl. No.:	17/795996
Filed (or PCT Filed):	January 26, 2021
PCT No.:	PCT/JP2021/002610
PCT Pub. No.:	WO2021/157427
PCT Pub. Date:	August 12, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20230073032 A1	Mar. 09, 2023

Foreign Application Priority Data

JP	2020-017945	Feb. 05, 2020
----	-------------	---------------

Publication Classification

Int. Cl.: H10N10/17 (20230101); H01M10/613 (20140101); H10N10/01 (20230101)

U.S. Cl.:

CPC H10N10/17 (20230201); H10N10/01 (20230201); H01M10/613 (20150401)

Field of Classification Search

CPC: H10N (10/17); H10N (10/01); H10N (10/00); H01M (10/613); H02J (1/08); H02J (7/345)

USPC: 320/135

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
9350173	12/2015	Kanayama	N/A	B60L 53/14
11258290	12/2021	Monden	N/A	H02J 7/04
2003/0227278	12/2002	Sakuragi	N/A	N/A
2010/0174417	12/2009	Iida	700/295	H01M 10/425
2012/0235490	12/2011	Lee	307/66	H02J 9/005
2014/0327450	12/2013	Schmauss	324/538	G01R 31/67
2017/0025878	12/2016	Tsai	N/A	H02J 7/0068

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
104081418	12/2013	CN	N/A
106851025	12/2016	CN	N/A
2004046088	12/2003	JP	N/A
2006149043	12/2005	JP	N/A
2010148234	12/2009	JP	N/A
2014092712	12/2013	JP	H02J 7/00
2016086578	12/2015	JP	N/A

OTHER PUBLICATIONS

Office Action in Chinese Appln. No. 202180011534.2, mailed on Apr. 9, 2025, 10 pages (with English translation). cited by applicant

Primary Examiner: Diao; M Baye

Attorney, Agent or Firm: Fish & Richardson P.C.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a National Stage application of International Application No.

PCT/JP2021/002610, filed on Jan. 26, 2021, which claims priority to Japanese Patent Application No. 2020-017945, filed on Feb. 5, 2020. The contents of the prior applications are incorporated herein in their entirety.

FIELD

(2) The present disclosure relates to a residual charge removal device.

BACKGROUND

(3) In the technical field related to electronic devices, a residual charge removal device as disclosed in Patent Literature 1 is known.

CITATION LIST

Patent Literature

(4) Patent Literature 1: JP 2014-092712 A

SUMMARY

Technical Problem

(5) A load such as an electronic device is driven by electric power supplied from a power supply device. In a state where electric power supply from the power supply device is stopped, if there is a residual charge, there is a possibility that the load malfunctions.

(6) An object of the present disclosure is to smoothly remove a residual charge in a state where electric power supply from the power supply device is stopped.

Solution to Problem

(7) According to an aspect of the present invention, a residual charge removal device comprises: a power storage unit; a first switching unit that is connected to a target to which electric power is supplied from a power supply device, and operates to remove a residual charge of the target by electric power supplied from the power storage unit; and a second switching unit that operates such that electric power is supplied from the power storage unit to the first switching unit when electric power supply to the target is stopped.

Advantageous Effects of Invention

(8) According to the present disclosure, it is possible to smoothly remove a residual charge in a state where electric power supply from the power supply device is stopped.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a diagram schematically illustrating a thermoelectric generator according to an embodiment.

(2) FIG. 2 is a perspective view schematically illustrating a thermoelectric power generation module according to the embodiment.

(3) FIG. 3 is a diagram schematically illustrating a power supply device according to the embodiment.

(4) FIG. 4 is a configuration diagram illustrating a main part of hardware of the thermoelectric generator according to the embodiment.

(5) FIG. 5 is a block diagram illustrating the thermoelectric generator according to the embodiment.

(6) FIG. 6 is a flowchart illustrating an operation of the thermoelectric generator according to the embodiment.

DESCRIPTION OF EMBODIMENTS

(7) Hereinafter, embodiments according to the present disclosure will be described with reference to the drawings, but the present disclosure is not limited thereto. The components of the embodiments described below can be appropriately combined. In addition, some components may not be used.

(8) [Thermoelectric Generator]

(9) FIG. 1 is a diagram schematically illustrating a thermoelectric generator 1 according to an embodiment. The thermoelectric generator 1 is installed in equipment B. The equipment B is provided in an industrial facility such as a factory, for example. As the equipment B, a motor that operates a pump is exemplified. The equipment B functions as a heat source of the thermoelectric generator 1.

(10) As illustrated in FIG. 1, the thermoelectric generator 1 includes a first housing 10, a second housing 20, a power supply device 30 including a thermoelectric power generation module 31, a microcomputer 40, a sensor 50, a wireless communication device 60, and a residual charge removal device 70.

(11) The first housing 10 accommodates the power supply device 30 and the residual charge removal device 70. The second housing 20 accommodates the microcomputer 40, the sensor 50, and the wireless communication device 60. Each of the first housing 10 and the second housing 20 is installed in the equipment B.

(12) The first housing 10 includes a heat receiving part 11, a heat dissipating part 12, and a peripheral wall part 13. An internal space of the first housing 10 is defined by the heat receiving part 11, the heat dissipating part 12, and the peripheral wall part 13. The power supply device 30 and the residual charge removal device 70 are disposed in the internal space of the first housing 10. An external space of the first housing 10 is an atmospheric space.

(13) The heat receiving part 11 is disposed so as to be in contact with the equipment B. The heat receiving part 11 is a plate-shaped member. The heat receiving part 11 is formed of a metal material such as aluminum or copper. The heat receiving part 11 receives heat from the equipment B. Heat of the heat receiving part 11 is transferred to the thermoelectric power generation module 31 through a heat transfer member 2.

(14) The heat dissipating part 12 faces the heat receiving part 11 with a gap interposed therebetween. The heat dissipating part 12 is a plate-shaped member. The heat dissipating part 12 is formed of a metal material such as aluminum or copper. The heat dissipating part 12 receives heat from the thermoelectric power generation module 31. Heat of the heat dissipating part 12 is released to the atmospheric space around the thermoelectric generator 1.

(15) The peripheral wall part 13 is disposed between a peripheral edge part of the heat receiving part 11 and a peripheral edge part of the heat dissipating part 12. The peripheral wall part 13 has an annular shape. The peripheral wall part 13 connects the heat receiving part 11 and the heat dissipating part 12. The peripheral wall part 13 is made of synthetic resin.

(16) The power supply device 30 functions as a power supply for the microcomputer 40, the sensor 50, and the wireless communication device 60. Each of the microcomputer 40, the sensor 50, and the wireless communication device 60 is driven on the basis of electric power supplied from the power supply device 30. Each of the microcomputer 40, the sensor 50, and the wireless communication device 60 is a load C that consumes electric power supplied from the power supply device 30.

(17) The thermoelectric generator 1 includes a power supply line 80 that transmits electric power from the power supply device 30 to the load C. In the embodiment, the power supply device 30 is connected to each of the microcomputer 40 and the sensor 50 via the power supply line 80.

(18) The power supply device 30 includes the thermoelectric power generation module 31. The thermoelectric power generation module 31 generates electric power using the Seebeck effect. The thermoelectric power generation module 31 is disposed between the heat receiving part 11 and the heat dissipating part 12. One end face 311 of the thermoelectric power generation module 31 is heated to give a temperature difference between one end face 311 and the other end face 312 of the thermoelectric power generation module 31, whereby the thermoelectric power generation module 31 generates electric power.

(19) The microcomputer 40 controls the thermoelectric generator 1. The microcomputer 40 is driven by electric power generated by the thermoelectric power generation module 31.

(20) The sensor **50** detects a state of the equipment B. Examples of the sensor **50** include a vibration sensor that detects vibration of the equipment B, and a temperature sensor that detects temperature of the equipment B. The sensor **50** is driven by electric power generated by the thermoelectric power generation module **31**.

(21) The wireless communication device **60** transmits detection data of the sensor **50**. The wireless communication device **60** is driven by electric power generated by the thermoelectric power generation module **31**. The detection data of the sensor **50** is transmitted to a management computer **100** existing outside the thermoelectric generator **1** by the wireless communication device **60**. The management computer **100** diagnoses the presence or absence of abnormality of the equipment B based on the detection data of the sensor **50**.

(22) The residual charge removal device **70** removes a residual charge of a target to which electric power is supplied from the power supply device **30**. In the embodiment, the target to which electric power is supplied from the power supply device **30** includes at least one of the load C that consumes electric power supplied from the power supply device **30**, and the power supply line **80** that transmits electric power from the power supply device **30** to the load C. The load C includes at least one of the microcomputer **40**, the sensor **50**, and the wireless communication device **60**.

(23) In a state where electric power supply from the power supply device **30** is stopped, there is a possibility that a residual charge is present in the target. The residual charge removal device **70** removes the residual charge from the target in the state where the electric power supply from the power supply device **30** is stopped.

(24) [Thermoelectric Power Generation Module]

(25) FIG. **2** is a perspective view schematically illustrating the thermoelectric power generation module **31** according to the embodiment. The thermoelectric power generation module **31** includes p-type thermoelectric semiconductor elements **31P**, n-type thermoelectric semiconductor elements **31N**, first electrodes **3**, second electrodes **4**, a first substrate **5**, and a second substrate **6**. The first electrodes **3** are disposed on the first substrate **5**. The second electrodes **4** are disposed on the second substrate **6**. On each of a surface of the first substrate **5** and a surface of the second substrate **6**, the p-type thermoelectric semiconductor elements **31P** and the n-type thermoelectric semiconductor elements **31N** are alternately arranged. Each of the first electrodes **3** is connected to each of the p-type thermoelectric semiconductor elements **31P** and the n-type thermoelectric semiconductor elements **31N**. Each of the second electrodes **4** is connected to each of the p-type thermoelectric semiconductor elements **31P** and the n-type thermoelectric semiconductor elements **31N**. One end of each of the p-type thermoelectric semiconductor elements **31P** and one end of each of the n-type thermoelectric semiconductor elements **31N** are connected to each of the first electrodes **3**. The other end of each of the p-type thermoelectric semiconductor elements **31P** and the other end of each of the n-type thermoelectric semiconductor elements **31N** are connected to each of the second electrodes **4**.

(26) Each of the p-type thermoelectric semiconductor elements **31P** and the n-type thermoelectric semiconductor elements **31N** includes, for example, a BiTe-based thermoelectric material. Each of the first substrate **5** and the second substrate **6** is formed of an electrically insulating material such as ceramics or polyimide.

(27) The first substrate **5** has an end surface **311**. The second substrate **6** has an end surface **312**. By heating the first substrate **5**, a temperature difference is provided between one end and the other end of each of the p-type thermoelectric semiconductor elements **31P** and the n-type thermoelectric semiconductor elements **31N**. When a temperature difference is given between one end and the other end of each of the p-type thermoelectric semiconductor elements **31P**, holes move in each of the p-type thermoelectric semiconductor elements **31P**. When a temperature difference is given between one end and the other end of each of the n-type thermoelectric semiconductor elements **31N**, electrons move in each of the n-type thermoelectric semiconductor elements **31N**. Each of the p-type thermoelectric semiconductor elements **31P** and each of the n-type thermoelectric

semiconductor elements **31N** are connected via each of the first electrodes **3** and each of the second electrodes **4**. A potential difference is generated between each of the first electrodes **3** and each of the second electrodes **4** by the holes and electrons. When the potential difference is generated between each of the first electrodes **3** and each of the second electrodes **4**, the thermoelectric power generation module **31** generates electric power. Lead wires **7** are connected to the first electrodes **3**. The thermoelectric power generation module **31** outputs electric power via the lead wires **7**.

(28) [Power Supply Device]

(29) FIG. **3** is a diagram schematically illustrating the power supply device **30** according to the embodiment. As illustrated in FIG. **3**, the power supply device **30** includes the thermoelectric power generation module **31**, a power supply switching unit **32**, and a power supply storage unit **33**.

(30) The thermoelectric power generation module **31** generates electric power using the Seebeck effect. Electric power generated by the thermoelectric power generation module **31** is boosted by a booster (not illustrated) and then supplied to the power supply switching unit **32**.

(31) The power supply switching unit **32** operates to switch between a power storage state in which electric power generated by the thermoelectric power generation module **31** is supplied to the power supply storage unit **33** and a supply state in which electric power stored in the power supply storage unit **33** is supplied to the load C.

(32) The power supply switching unit **32** monitors the power storage state of the power supply storage unit **33**, and switches from the power storage state to the supply state when a power storage amount of the power supply storage unit **33** reaches a specified value.

(33) The power supply switching unit **32** is connected to the power supply storage unit **33** via an electric line **82**. The power supply switching unit **32** is connected to the load C via the power supply line **80**. Further, the power supply switching unit **32** is connected to the microcomputer **40** via a control line **90**. The control line **90** transmits a control signal from the microcomputer **40** to the power supply device **30**. The control signal output from the microcomputer **40** is transmitted to the power supply switching unit **32** via the control line **90**. The power supply switching unit **32** is controlled by the microcomputer **40**.

(34) When the thermoelectric power generation module **31** generates electric power, as indicated by an arrow R1, the power supply switching unit **32** operates such that the electric power generated by the thermoelectric power generation module **31** is supplied to the power supply storage unit **33** through the electric line **82**. The power supply storage unit **33** stores electric power generated by the thermoelectric power generation module **31**.

(35) When the power storage amount of the power supply storage unit **33** becomes greater than or equal to the specified value, the power supply storage unit **33** releases electric power. As indicated by an arrow R2, the electric power released from the power supply storage unit **33** is supplied to the load C via the power supply line **80**. The load C consumes the electric power supplied from the power supply storage unit **33**.

(36) After the power supply storage unit **33** releases the electric power, as indicated by the arrow R1, the power supply switching unit **32** operates such that the electric power generated by the thermoelectric power generation module **31** is supplied to the power supply storage unit **33** via the electric line **82**. The power supply storage unit **33** stores the electric power generated by the thermoelectric power generation module **31** again.

(37) When the power storage amount of the power supply storage unit **33** becomes greater than or equal to the specified value, the power supply storage unit **33** releases electric power. As indicated by the arrow R2, the electric power released from the power supply storage unit **33** is supplied to the load C again via the power supply line **80**. The load C consumes the electric power supplied from the power supply storage unit **33**.

(38) In this way, a power storage state in which the electric power generated by the thermoelectric power generation module **31** is supplied to the power supply storage unit **33**, and a supply state in

which the electric power stored in the power supply storage unit **33** is supplied to the load C are repeated. The power supply storage unit **33** performs the power storage intermittently. Electric power is intermittently supplied to the load C.

(39) In the supply state, the thermoelectric power generation module **31**, the power supply storage unit **33**, and the load C are connected.

(40) In the power storage state, the supply of electric power from the power supply device **30** (power supply storage unit **33**) to the load C is stopped. That is, in the power storage state, the load C is turned off.

(41) When there is a residual charge in at least a part of the load C, the power supply line **80**, and the control line **90** after the supply of electric power from the power supply device **30** to the load C is stopped and the load C is turned off, there is a possibility that the load C malfunctions.

(42) In the embodiment, the residual charge removal device **70** operates to remove the residual charge when switched from the supply state to the power storage state.

(43) [Residual Charge Removal Device]

(44) FIG. **4** is a configuration diagram illustrating a main part of hardware of the thermoelectric generator **1** according to the embodiment. As illustrated in FIG. **4**, the thermoelectric generator **1** includes the thermoelectric power generation module **31**, a DC/DC converter **320**, a power supply capacitor **330**, the microcomputer **40**, and the residual charge removal device **70**.

(45) The DC/DC converter **320** is connected to each of the thermoelectric power generation module **31**, the power supply capacitor **330**, and the microcomputer **40**. The DC/DC converter **320** is disposed between the power supply capacitor **330** and the microcomputer **40**. The DC/DC converter **320** is connected to the power supply capacitor **330** via the electric line **82**. The DC/DC converter **320** is connected to the microcomputer **40** via the power supply line **80**.

(46) The DC/DC converter **320** includes a switch **321** and a switch control terminal **322**.

(47) The switch **321** switches between an energizable state in which the power supply capacitor **330** and the microcomputer **40** can be energized and a non-energizable state in which the power supply capacitor **330** and the microcomputer **40** cannot be energized. When the switch **321** is turned on, the power supply capacitor **330** and the microcomputer **40** are energizable. When the switch **321** is turned off, the power supply capacitor **330** and the microcomputer **40** are not energized.

(48) The switch control terminal **322** controls the switch **321**. The switch control terminal **322** is controlled by the microcomputer **40**. The switch control terminal **322** is set to either a High level or a Low level. When the switch control terminal **322** is at the High level, an on state of the switch **321** is maintained.

(49) The power supply capacitor **330** stores electric power generated by the thermoelectric power generation module **31**. When a power storage amount of the power supply capacitor **330** is less than a specified value, the switch **321** is turned off. When the power storage amount of the power supply capacitor **330** is greater than or equal to the specified value, the switch **321** is turned on.

(50) When the switch **321** is turned on, the power supply capacitor **330** and the microcomputer **40** are energizable. When the power supply capacitor **330** and the microcomputer **40** are in the energizable state, electric power stored in the power supply capacitor **330** is supplied to the microcomputer **40** via the DC/DC converter **320** and the power supply line **80**.

(51) The microcomputer **40** includes a micro processing unit (MPU). The microcomputer **40** is driven on the basis of the electric power supplied from the power supply capacitor **330**. The microcomputer **40** being in a driving state means that the electric power is being supplied to the microcomputer **40**. The microcomputer **40** being in a stopped state means that no electric power is supplied to the microcomputer **40**.

(52) The microcomputer **40** includes a busy terminal **440**. When the microcomputer **40** is in the driving state, the busy terminal **440** is set to a High level. When the microcomputer **40** is in the stopped state, the busy terminal **440** is set to a Low level.

(53) By setting the busy terminal **440** to the High level, the switch control terminal **322** is set to the High level, and the on state of the switch **321** is maintained.

(54) The residual charge removal device **70** includes a capacitor **710**, a first transistor **720**, and a second transistor **730**.

(55) A diode **D1** is disposed between the busy terminal **440** and the capacitor **710**. By setting the busy terminal **440** to the High level, electric power is supplied from the power supply capacitor **330** to the capacitor **710** via the microcomputer **40**. That is, when the busy terminal **440** is set to the High level, the capacitor **710** is stored. By setting the busy terminal **440** to the Low level, the busy terminal **440** is connected to a ground GND.

(56) The first transistor **720** is connected to the power supply line **80**. A diode **D3** is disposed on the power supply line **80**. The first transistor **720** is a field effect transistor (FET). In the embodiment, the first transistor **720** is an N-channel metal oxide semiconductor field effect transistor (N-channel MOSFET).

(57) When electric power is supplied from the capacitor **710** to the first transistor **720**, the first transistor **720** is turned on. When the first transistor **720** is turned on, the power supply line **80** and the ground GND are energizable via the first transistor **720**. When the power supply line **80** and the ground GND are in the energizable state, the residual charge of the power supply line **80** is released to the ground GND. That is, the residual charge of the power supply line **80** is removed.

(58) The second transistor **730** is connected to the microcomputer **40**. A diode **D2** is disposed between the second transistor **730** and the ground GND. The second transistor **730** is a field effect transistor (FET). In the embodiment, the second transistor **730** is a P-channel metal oxide semiconductor field effect transistor (P-channel MOSFET). When the busy terminal **440** is set to the High level, the second transistor **730** is turned off. By setting the busy terminal **440** to the Low level, the second transistor **730** is turned on. That is, the second transistor **730** is turned off when the microcomputer **40** is in the driving state, and is turned on when the microcomputer **40** is in the stopped state.

(59) When the second transistor **730** is turned on, the capacitor **710** and the first transistor **720** can be energized via the second transistor **730**. When the capacitor **710** and the first transistor **720** are in the energizable state, the electric power (charge) stored in the capacitor **710** is supplied to the first transistor **720**. When electric power is supplied from the capacitor **710** to the first transistor **720**, the first transistor **720** is turned on.

(60) FIG. 5 is a block diagram illustrating the thermoelectric generator **1** according to the embodiment. As illustrated in FIG. 5, the thermoelectric generator **1** includes the power supply device **30**, the microcomputer **40**, the sensor **50**, the wireless communication device **60**, and the residual charge removal device **70**.

(61) The power supply device **30** includes the thermoelectric power generation module **31**, the power supply switching unit **32**, and the power supply storage unit **33**.

(62) The power supply switching unit **32** includes the DC/DC converter **320** described with reference to FIG. 4. The power supply switching unit **32** operates to switch between the power storage state in which electric power generated by the thermoelectric power generation module **31** is supplied to the power supply storage unit **33** and the supply state in which electric power stored in the power supply storage unit **33** is supplied to the target.

(63) The power supply switching unit **32** operates so as to be in the power storage state when the power storage amount of the power supply storage unit **33** is less than the specified value, and to be in the supply state when the power storage amount of the power supply storage unit **33** is greater than or equal to the specified value. When the power storage amount of the power supply storage unit **33** changes from a value less than the specified value to a value greater than or equal to the specified value, the switch **321** of the power supply switching unit **32** is turned on. When the switch **321** is turned on, the power supply storage unit **33** and the microcomputer **40** can be energized.

(64) The microcomputer **40** includes a micro processing unit (MPU). The microcomputer **40** includes a detection data acquisition unit **41**, a communication control unit **42**, a power supply control unit **43**, and a removal control unit **44**.

(65) The detection data acquisition unit **41** acquires detection data of the sensor **50**. The communication control unit **42** controls the wireless communication device **60**. The communication control unit **42** controls the wireless communication device **60** so that the detection data of the sensor **50** is transmitted to the management computer **100**.

(66) The power supply control unit **43** outputs a control signal for controlling the power supply switching unit **32**. The control line **90** that transmits the control signal from the microcomputer **40** to the power supply device **30** is provided. The control signal output from the power supply control unit **43** is transmitted to the power supply switching unit **32** via the control line **90**.

(67) The removal control unit **44** outputs a control signal for controlling the residual charge removal device **70**.

(68) The residual charge removal device **70** includes a power storage unit **71**, a first switching unit **72**, and a second switching unit **73**.

(69) The power storage unit **71** stores electric power (charge). The power storage unit **71** includes the capacitor **710** described with reference to FIG. **4**. The power storage unit **71** stores electric power supplied from the power supply device **30** (power supply storage unit **33**) via the power supply line **80** and the microcomputer **40**.

(70) The first switching unit **72** is connected to the target to which electric power is supplied from the power supply device **30**. In the embodiment, the first switching unit **72** is connected to each of the power supply line **80** and the control line **90**. The first switching unit **72** is connected to the power supply line **80** via a connection line **74**. The first switching unit **72** is connected to the control line **90** via a connection line **75**.

(71) The first switching unit **72** operates such that a residual charge of the power supply line **80** and a residual charge of the control line **90** are removed by the supply of electric power from the power storage unit **71**. The first switching unit **72** is connected to the ground GND. When electric power is supplied from the power storage unit **71**, the first switching unit **72** operates so that the residual charge of the power supply line **80** and the residual charge of the control line **90** are released to the ground GND.

(72) The first switching unit **72** includes the first transistor **720** described with reference to FIG. **4**. When electric power is supplied from the power storage unit **71**, the first switching unit **72** is turned on to connect each of the power supply line **80** and the control line **90** to the ground GND. When the first switching unit **72** is turned on, the residual charge of the power supply line **80** and the residual charge of the control line **90** are released to the ground GND via the first switching unit **72**.

(73) The second switching unit **73** operates such that electric power is supplied from the power storage unit **71** to the first switching unit **72** when electric power supply from the power supply device **30** to the target is stopped. In the embodiment, the second switching unit **73** operates such that electric power is supplied from the power storage unit **71** to the first switching unit **72** when electric power supply from the power supply device **30** to the microcomputer **40** and the sensor **50** is stopped.

(74) As described above, the power supply switching unit **32** switches between the power storage state in which electric power generated by the thermoelectric power generation module **31** is supplied to the power supply storage unit **33**, and the supply state in which electric power stored in the power supply storage unit **33** is supplied to the microcomputer **40** and the sensor **50**. The second switching unit **73** operates such that electric power is supplied from the power storage unit **71** to the first switching unit **72** when the supply state is switched to the power storage state.

(75) The second switching unit **73** includes the second transistor **730** described with reference to FIG. **4**. When electric power supply from the power supply storage unit **33** to the microcomputer

40 is stopped, the second switching unit **73** is turned on to connect the power storage unit **71** and the first switching unit **72**. When the second switching unit **73** is turned on, electric power of the power storage unit **71** is supplied to the first switching unit **72** via the second switching unit **73**. The first switching unit **72** is turned on by electric power supplied from the power storage unit **71**. When the first switching unit **72** is turned on, the residual charge of the power supply line **80** and the residual charge of the control line **90** are released to the ground GND via the first switching unit **72**.

(76) [Operation]

(77) FIG. **6** is a flowchart illustrating an operation of the thermoelectric generator **1** according to the embodiment. Electric power generated by the thermoelectric power generation module **31** is supplied to the power supply storage unit **33**. The power supply storage unit **33** stores the electric power generated by the thermoelectric power generation module **31** (Step S1).

(78) When the power storage amount of the power supply storage unit **33** is less than the specified value, the switch control terminal **322** of the power supply switching unit **32** is at the Low level, and the switch **321** of the power supply switching unit **32** is in an off state. When the power storage amount of the power supply storage unit **33** is less than the specified value, the power storage state in which the electric power generated by the thermoelectric power generation module is supplied to the power supply storage unit **33** is maintained.

(79) When the power storage amount of the power supply storage unit **33** changes from a value less than the specified value to a value greater than or equal to the specified value, the power storage state is switched to the supply state. That is, when the power storage amount of the power supply storage unit **33** is greater than or equal to the specified value, the switch **321** of the power supply switching unit **32** is turned on, and the electric power stored in the power supply storage unit **33** is supplied to the load C including the microcomputer **40** and the sensor **50** (Step S2).

(80) When electric power is supplied to the microcomputer **40**, the busy terminal **440** of the microcomputer **40** is set to the High level. By setting the busy terminal **440** to the High level, the switch control terminal **322** of the power supply switching unit **32** is set to the High level, and the on state of the switch **321** of the power supply switching unit **32** is maintained.

(81) When electric power is supplied to the microcomputer **40** and the sensor **50**, each of the microcomputer **40** and the sensor **50** is driven (Step S3).

(82) Each of the microcomputer **40** and the sensor **50** performs specified processing. The sensor **50** detects a state of the equipment B. The detection data acquisition unit **41** acquires detection data of the sensor **50**. The communication control unit **42** controls the wireless communication device **60** to transmit the detection data of the sensor **50** to the management computer **100**.

(83) Further, when the microcomputer **40** is driven, electric power from the power supply device **30** is supplied to the power storage unit **71** via the microcomputer **40**. The power storage unit **71** stores the electric power supplied from the power supply device **30** via the microcomputer **40** (Step S4).

(84) When the processing of each of the microcomputer **40** and the sensor **50** ends, the power supply control unit **43** outputs a control signal to the power supply switching unit **32** so as to switch from the supply state to the power storage state. The control signal output from the power supply control unit **43** is transmitted to the power supply switching unit **32** via the control line **90** (Step S5).

(85) In the embodiment, when the processing of each of the microcomputer **40** and the sensor **50** ends, the busy terminal **440** is set to the Low level. When the busy terminal **440** is set to the Low level, a control signal for setting the switch control terminal **322** to the Low level is output from the power supply control unit **43** to the power supply switching unit **32**. The control signal for setting the switch control terminal **322** to the Low level corresponds to a control signal for switching from the supply state to the power storage state. When the switch control terminal **322** is set to the Low level, the switch **321** of the power supply switching unit **32** is turned off to switch from the supply state to the power storage state.

- (86) In the power storage state, the supply of electric power to the microcomputer **40** and the sensor **50** is stopped. The microcomputer **40** and the sensor **50** are turned off (Step S6).
- (87) When the supply of electric power to the microcomputer **40** and the sensor **50** is stopped, the second switching unit **73** operates so that electric power is supplied from the power storage unit **71** to the first switching unit **72** (Step S7).
- (88) In the embodiment, when the busy terminal **440** is set to the Low level, the second switching unit **73** is turned on. When the second switching unit **73** is turned on, electric power of the power storage unit **71** is supplied to the first switching unit **72** via the second switching unit **73**. When the electric power of the power storage unit **71** is supplied to the first switching unit **72**, the first switching unit **72** is turned on (Step S8).
- (89) When the first switching unit **72** is turned on, the power supply line **80** and the ground GND are energizable via the first switching unit **72**. As a result, a residual charge existing in the power supply line **80** is released to the ground GND via the first switching unit **72**. That is, the residual charge of the power supply line **80** is removed (Step S9).
- (90) The processing from Step S1 to Step S9 is repeated. That is, the operation of storing power in the power supply storage unit **33** with the electric power generated by the thermoelectric power generation module **31**, the operation of supplying the electric power stored in the power supply storage unit **33** to the load C, and the operation of removing the residual charge are repeated.

Advantageous Effects

- (91) As described above, according to the embodiment, the residual charge removal device **70** smoothly removes the residual charge existing in the target in a state where the supply of electric power from the power supply device **30** to the target is stopped. When the supply of electric power from the power supply device **30** to the target is stopped, the second switching unit **73** is operated such that electric power stored in the power storage unit **71** is supplied to the first switching unit **72**. As a result, even if the supply of electric power from the power supply device **30** is stopped, the first switching unit **72** can be turned on based on the electric power supplied from the power storage unit **71**. When the first switching unit **72** is turned on, the residual charge of the target is released to the ground GND via the first switching unit **72**. Therefore, the residual charge present in the target is smoothly removed.
- (92) The power storage unit **71** stores electric power supplied from the power supply device **30** via the microcomputer **40**. As a result, the electric power from the power supply device **30** is efficiently stored in the power storage unit **71**.
- (93) The power storage state in which the electric power generated by the thermoelectric power generation module **31** is supplied to the power supply storage unit **33**, and the supply state in which the electric power stored in the power supply storage unit **33** is supplied to the load C are repeated. In the embodiment, the power supply storage unit **33** intermittently performs power storage, and the detection data of the sensor **50** is intermittently transmitted. When the supply state is switched to the power storage state, the residual charge is removed by the residual charge removal device **70**. Accordingly, malfunction of the sensor **50** and malfunction of the wireless communication device **60** are suppressed. Therefore, the detection data of the state of the equipment B is appropriately transmitted to the management computer **100**.

Other Embodiments

- (94) In the above-described embodiment, the first switching unit **72** is connected to each of the power supply line **80** and the control line **90**. As a result, the residual charge of the power supply line **80** and the residual charge of the control line **90** are smoothly removed. Note that the first switching unit **72** may be connected to the power supply line **80** but not to the control line **90**. Further, the first switching unit **72** may be connected to the load C. That is, the first switching unit **72** may be connected to at least one of the microcomputer **40**, the sensor **50**, and the wireless communication device **60**.
- (95) In the above-described embodiment, the power storage unit **71** stores the electric power

supplied from the power supply device **30** via the microcomputer **40**. The electric power output from the power supply device **30** may be supplied to the power storage unit **71** without passing through the microcomputer **40**.

Claims

1. A residual charge removal device comprising: a power storage unit; a first switching unit that is connected to a target to which electric power is supplied from a power supply device, and operates to remove a residual charge of the target by electric power supplied from the power storage unit; and a second switching unit that operates such that electric power is supplied from the power storage unit to the first switching unit when electric power supply to the target is stopped.
 2. The residual charge removal device according to claim 1, wherein the power storage unit stores electric power supplied from the power supply device via the target.
 3. The residual charge removal device according to claim 1, wherein the target includes at least one of a load that consumes electric power supplied from the power supply device, and a power supply line that transmits electric power from the power supply device to the load.
 4. The residual charge removal device according to claim 1, wherein the target includes a computer that consumes electric power supplied from the power supply device.
 5. The residual charge removal device according to claim 4, wherein a control line that transmits a control signal from the computer to the power supply device is provided, and the first switching unit operates to remove a residual charge in the control line.
 6. The residual charge removal device according to claim 1, wherein; the power supply device includes a thermoelectric power generation module, a power supply switching unit, and a power supply storage unit, the power supply switching unit operates to switch between a power storage state in which electric power generated by the thermoelectric power generation module is supplied to the power supply storage unit and a supply state in which electric power stored in the power supply storage unit is supplied to the target, and the second switching unit operates such that electric power is supplied from the power storage unit to the first switching unit when the supply state is switched to the power storage state.
-